TrustStack

A Local-First, Evidence-Grounded Evaluation Standard for Trustworthy AI

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1. Introduction and Problem Setting

TrustStack starts from a simple problem: a fluent answer is not enough when a user must decide whether an AI output is safe, supported, and auditable before using it.

Research question

How can a user judge whether an AI answer is trustworthy when conventional chat interfaces expose fluency but hide evidence quality, contradiction risk, and review conditions?

Project thesis

TrustStack reframes trust as an evidence-grounded evaluation task. It turns trust into a structured review artifact built from evidence, scoring, risk signals, and explanation.

Core task

Grounded trust review

Primary risk

Unsupported confidence

Decision basis

Evidence + diagnostics

2. Related Work and Contribution

TrustStack sits between retrieval-augmented generation, hallucination analysis, and structured evaluation frameworks. The contribution is not a new base model. It is a new evidence-first evaluation layer around existing generation pipelines.

RAG systems

Model outputs should be conditioned on retrieved evidence rather than unconstrained recall.

Reliability research

Plausible answers can still be unsupported, unsafe, or operationally misleading.

Evaluation frameworks

Reproducible prompts, dimensions, judgments, and exportable artifacts matter.

TrustStack contribution

TrustStack integrates these threads into one local-first workflow centered on answer inspection rather than answer appearance.

3. Data

The evaluation uses two data families: live evidence uploaded by the user and benchmark-oriented corpora used to stress the system under reproducible conditions.

Live evidence path

- PDF, DOCX, TXT, and Markdown uploads
- Parsed, chunked, indexed, and stored in MongoDB
- Used for real-time evidence-grounded question answering

Benchmark path

- Seven synthetic stress-test packets
- Checked-in public benchmark subsets: SciFact and HotpotQA
- Same evaluation protocol applied across both paths

User uploads | Synthetic stress tests | External benchmarks

4. Methods and System Architecture

TrustStack is implemented as a five-layer pipeline so each stage can be validated independently and each failure can be localized.

Pipeline

1. Ingest
2. Retrieve
3. Generate
4. Evaluate
5. Review / Export

Implementation

- MongoDB-backed documents, chunks, and run history
- Retrieval via vector or lexical fallback
- Grounded answer generation with deterministic fallback mode

Verification stack

- Backend unit tests
- Real Mongo integration tests
- Browser E2E tests
- Served-stack smoke verification

5. TrustStack Evaluation Standard v2.0

The TrustStack Evaluation Standard converts one answer into an evidence-first review packet rather than a single opaque score.

Ten scored dimensions

Retrieval relevance, evidence sufficiency, citation traceability, claim support, contradiction risk, completeness, abstention behavior, answer discipline, safety risk, and calibration.

Verdicts

Pass / Review / Fail

Diagnostics

Claims + checks + flags

Output

Reproducible report artifact

Why this matters

6. Synthetic Benchmark Findings

The synthetic benchmark tests whether TrustStack reacts sensibly when the evidence environment changes under controlled conditions.

Synthetic evaluation design

Seven corpus conditions stress aligned evidence, contradiction, sparse evidence, lexical drift, unsafe procedural advice, and consensus-style multi-document overlap.

Interpretation

The benchmark does not claim absolute trustworthiness. It tests whether scores move in the expected direction when evidence quality changes.

Best synthetic score

76.44 / 100

Worst synthetic score

59.97 / 100

Weakest category

Grounding + retrieval

7. External Benchmark Findings

To reduce the strongest threat to validity, TrustStack was also evaluated on checked-in public benchmark subsets from SciFact and HotpotQA.

What improved

The paper no longer relies only on internally authored synthetic corpora. External benchmark slices test the score and verdict logic beyond internal packets.

What remains weak

The current local runtime still shows a substantial calibration gap: citation alignment is relatively strong, but task-level correctness is weak.

Aggregate TrustStack score

63.09 / 100

Task metric

0.141

Citation alignment

83.8%

8. Threats to Validity and Limitations

A defensible AI project must separate what has been shown from what still requires stronger evidence.

Current limits

- Local lexical retrieval remains a bottleneck.
- Extractive fallback generation is reproducible but not state-of-the-art.
- No formal human analyst study has been completed yet.

Why the project is still credible

- The limitations are explicit, measured, and tied to concrete next experiments.
- The paper now includes real external benchmarks, not just synthetic data.
- The evaluation and export stack is reproducible end to end.

9. Conclusion

TrustStack contributes a reusable workflow for grounded evaluation, benchmarking, and reporting.

Main contribution

TrustStack is valuable not because it makes AI look more persuasive, but because it makes trust inspectable. It connects live evidence review, a formal evaluation standard, synthetic benchmarking, public-benchmark evaluation, and exportable IEEE-ready artifacts into one coherent system.

System role
Evidence-first evaluator

Empirical basis
Synthetic + external

Deliverable
Report-ready artifacts

Questions